Unit 1 Progress Check: MCQ Part A

1. A student wants to create an algorithm to calculate the average (mean) of two quiz scores. Which of the following algorithms can be used to determine the correct average?

(A)

Step one:	Add the two quiz scores.
Step two:	Multiply the result of step one by 2 to determine the average.

(B)

Step one:	Add the two quiz scores.
Step two:	Divide the result of step one by 2 to determine the average.

(C)

Step one:	Subtract the lower quiz score from the higher quiz score.
Step two:	Multiply the result of step one by 2 to determine the average.

(D)

Step one:	Subtract the lower quiz score from the higher quiz score.
Step two:	Divide the result of step one by 2 to determine the average.

2. A student wants to create an algorithm to calculate the combined total area of two rectangles. Which of the following algorithms can be used to perform this calculation?

(A)

Step 1:	Multiply the length of the first rectangle by the length of the second rectangle.
Step 2:	Multiply the width of the first rectangle by the width of the second rectangle.
Step 3:	Add the result of step 1 to the result of step 2.

(B)

Step 1:	Multiply the length of the first rectangle by the width of the first rectangle.
Step 2:	Multiply the length of the second rectangle by the width of the second rectangle.
Step 3:	Add the result of step 1 to the result of step 2.

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(C)

Step 1:	Add the length of the first rectangle to the width of the first rectangle.
Step 2:	Add the length of the second rectangle to the width of the second rectangle.
Step 3:	Multiply the result of step 1 by the result of step 2.

(D)

Step 1:	Add the length of the first rectangle to the length of the second rectangle.
Step 2:	Add the width of the first rectangle to the width of the second rectangle.
Step 3:	Multiply the result of step 1 by the result of step 2.

3. A student wrote a program to calculate the average of several numeric scores. The program compiles without error but produces incorrect results when it runs. The student suspects there is a logic error in the code.

Which of the following strategies is most likely to help identify the logic error?

(A) Changing random parts of the code and rerunning the program until the error is resolved
(B) Inserting print statements throughout the code to check the calculations at various points in the execution
(C) Relying on the compiler's error messages to detect the logic error
(D) Running the program multiple times without making changes to see if the error corrects itself

4. Consider the following variable declaration.

```
int x;
```

Which of the following values can be stored in the variable x?

(A)
0.5

(B)
1

(C)
true

(D)
false

5. Consider the following variable declaration.

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boolean b;

Which of the following values can be stored in the variable b?

(A)
-1

(B)
0

(C)
1.0

(D)
false

6. A code segment will use variables to represent a student's age, in years, and whether the student has a driver's license. Which of the following variable declarations are most appropriate for the code segment?

(A)
```
boolean age;
double hasLicense;
```

(B)
```
boolean age;
boolean hasLicense;
```

(C)
```
int age;
boolean hasLicense;
```

(D)
```
int age;
double hasLicense;
```

7. Consider the following code segment.

```
int a = 1;
int b = 2;
int c = 3;
int d = 4;
double x = a + b * c % d;
```

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What is the value of x after executing the code segment?

(A)
1.0

(B)
2.5

(C)
3.0

(D)
7.0

8. Which of the following code segments will print the word Hello?

(A)
System.out.print("Hello");

(B)
System.out.print(Hello);

(C)
"System.out.print(Hello)";

(D)
System.out.print(He);
System.out.print(llo);

9. Consider the following statement.

System.out.print("AP \n Computer \n Science \n A \n rocks");

Which of the following best describes the behavior of this statement?

(A) It prints five lines of text, each separated by a line break indicated by the escape sequence \n.
(B) It prints four lines of text because the escape sequence \n appears four times in the string literal.
(C) It prints one line of text because the printed value is a single string literal.
(D) It prints nothing due to a syntax error.

10. Consider the following code segment.

```
boolean a = true;
boolean b = false;
```

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```
boolean temp = a;
a = b;
b = temp;
System.out.println(a);
System.out.println(b);
```

What is printed as a result of executing this code segment?

(A)

true

true

(B)

true

false

(C)

false

true

(D)

false

false

11. The following code segment contains an error.

```
double pi = 3.14;    // line 1
pi = 3.14159;        // line 2
double other = null; // line 3
double example;      // line 4
example = pi;        // line 5
```

Which of the following best describes the error in this code segment?

(A) In line 2, pi is assigned a value after it was already assigned a value in line 1.
(B) In line 3, null cannot be assigned to a non-reference type.
(C) In line 4, example should be assigned a value.
(D) In line 5, a variable cannot be assigned to another variable.

12. The volume of a cylinder is equal to the height times the area of the circular base. The area of the circular base is equal to π (pi) times the square of the radius.

The following code segment is intended to compute and print the volume of a cylinder with radius r and height h. Assume that the double variables r, h, and pi have been properly declared and initialized.

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```
/* missing code */
System.out.print(volume);
```

Which of the following **cannot** be used to replace /* missing code */ so that the code segment works as intended?

(A)

```
double baseArea;
double volume;
pi * r * r = baseArea;
baseArea * h = volume;
```

(B)

```
double baseArea = pi * r * r;
double volume = baseArea * h;
```

(C)

```
double volume = pi * r * r ;
volume = volume * h;
```

(D)

```
double volume = pi * r * r * h;
```